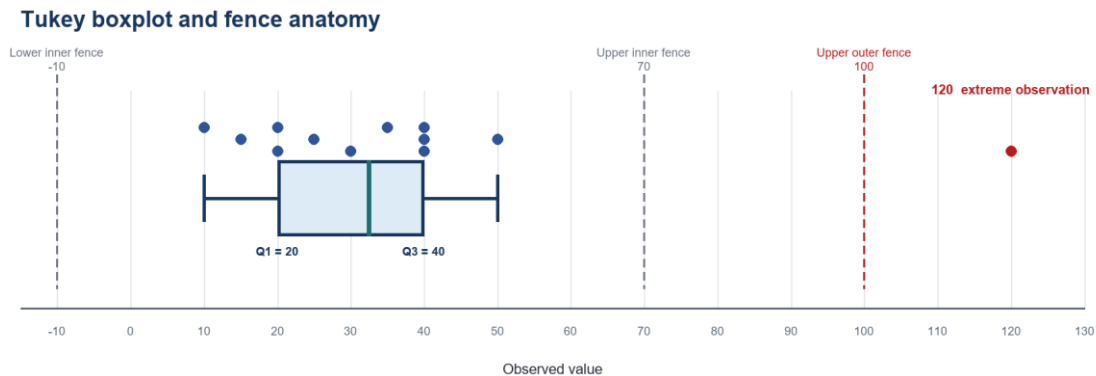


# How to Detect and Handle Outliers in Python

A practical guide for data science students using IQR and Z scores with safer data checks

By Mohamed Hisham, Anas, and Mahmoud



*A boxplot gives a quick view of the middle of the data and values far from it*

## 2 The Problem

An outlier is a value that sits far from most of the data. It might be a typing mistake, a sensor problem, or a rare but correct observation. Outliers matter because they can move the average, increase the standard deviation, and change a model's results.

The common mistake is to delete every value that a rule flags. A detection rule only tells us what to review. We still need to check where the value came from and decide whether to keep it, correct it, or use another treatment.

### What you will learn

- How to find possible outliers with the interquartile range
- When a classical Z score is useful and when it can miss a problem
- How a modified Z score can handle extreme values more safely
- How to review and treat a flagged value without losing the raw data

## 3 Concept and Architecture

### A flag is not a final answer

Outlier detection and outlier treatment are two different jobs. Detection creates a shortlist. Treatment happens only after we check the record, its units, and the reason it looks unusual.

A good process keeps the original value. If we create a corrected, capped, or transformed value, we store it in a new column so another person can reproduce the analysis.

## The IQR method

The interquartile range uses the middle half of the data. Q1 is the 25th percentile and Q3 is the 75th percentile. Their difference is the IQR.

$$\text{IQR} = Q_3 - Q_1$$

A common rule flags values below Q1 minus 1.5 IQR or above Q3 plus 1.5 IQR. Because the rule uses quartiles, one very large value does not move the limits as much as it moves the mean and standard deviation.

$$\text{LF} = Q_1 - 1.5 \times \text{IQR} \quad \text{UF} = Q_3 + 1.5 \times \text{IQR}$$

## The classical Z score

A Z score measures how many standard deviations a value is from the mean. Analysts often review values with an absolute Z score above 3. This guideline works best when the distribution is roughly normal and a few extremes have not distorted the mean and standard deviation.

$$Z_i = (x_i - \bar{x}) / s$$

## The modified Z score

The modified Z score replaces the mean with the median and the standard deviation with the median absolute deviation, or MAD. These values are less affected by a small number of extreme observations. A common review point is an absolute modified Z score above 3.5.

$$M_i = 0.6745 (x_i - \tilde{x}) / \text{MAD} \quad \text{MAD} = \text{median}(|x_i - \tilde{x}|)$$

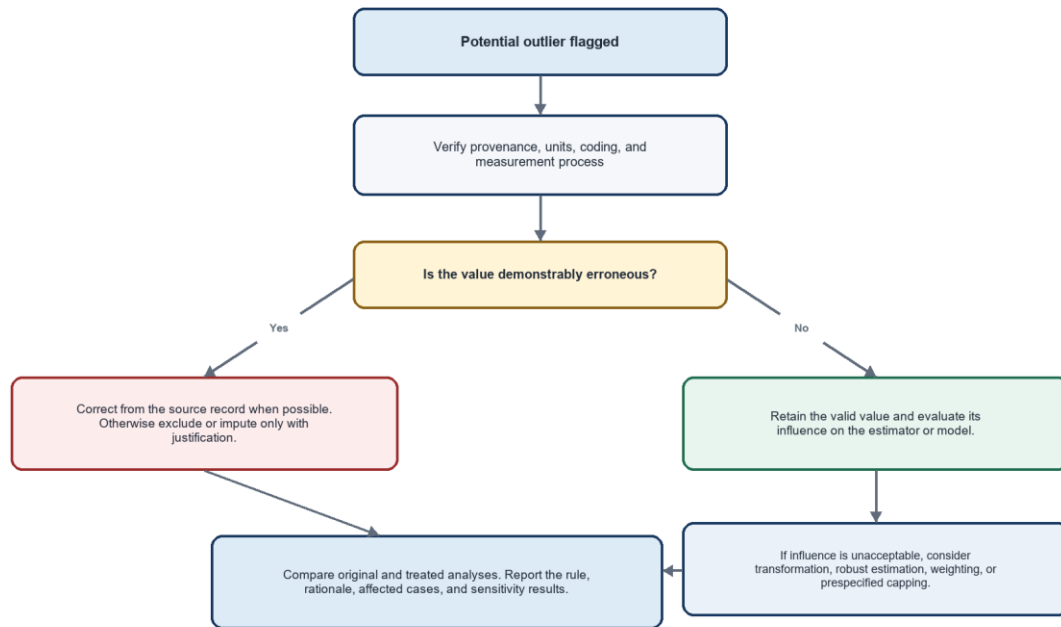
## Which method should you start with

| Method     | Starting rule            | Good starting point                  | Main warning                           |
|------------|--------------------------|--------------------------------------|----------------------------------------|
| IQR        | Outside 1.5 IQR fences   | Skewed data and quick exploration    | Can flag values in a valid long tail   |
| Z score    | Absolute Z above 3       | Stable data that is close to normal  | Extreme values can move its own limits |
| Modified Z | Absolute score above 3.5 | Data with several suspected extremes | MAD can be zero when many values match |

## A simple decision workflow

The workflow below starts with a statistical flag. It then checks the source of the value before any change is made.

### Evidence based outlier decision framework



*A flagged value should be checked before it is corrected, removed, or changed*

## 4 Step by Step Implementation

We will use a small dataset. Most values are close to 50, while 180 and 200 are far from the main group. The example is simple enough to follow, but the same steps work with a DataFrame column.

### Step 1 Create the data

```
import numpy as np
import pandas as pd

values = [48, 50, 49, 51, 52, 47, 50, 49, 51, 50, 180, 200]
df = pd.DataFrame({"value": values})
```

We keep the source values in one column. Any later treatment will go into a separate column.

### Step 2 Explore before applying a rule

```
print(df["value"].describe())
df["value"].plot(kind="box", title="Value distribution")
```

The summary shows the center and spread. The boxplot makes the two large values easy to see. A plot also helps us notice skewness, clusters, or data-entry limits that a single score cannot explain.

### Step 3 Add the IQR flag

```
q1 = df["value"].quantile(0.25)
q3 = df["value"].quantile(0.75)
iqr = q3 - q1

lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

df["iqr_outlier"] = ~df["value"].between(lower, upper)
```

Pandas calculates Q1 and Q3, then the code builds the two fences. The Boolean column keeps the flag separate from the original measurement.

## Step 4 Add the classical Z score

```
mean = df["value"].mean()
std = df["value"].std(ddof=1)

df["z_score"] = (df["value"] - mean) / std
df["z_outlier"] = df["z_score"].abs() > 3
```

The code uses the sample standard deviation. In this example, 180 and 200 increase both the mean and the standard deviation. That makes their Z scores smaller than we might expect.

## Step 5 Add the modified Z score

```
median = df["value"].median()
mad = np.median(np.abs(df["value"] - median))

if mad == 0:
    df["modified_z"] = np.nan
else:
    df["modified_z"] = 0.6745 * (df["value"] - median) / mad

df["modified_z_outlier"] = df["modified_z"].abs() > 3.5
```

The zero check matters. If many values are identical, MAD can be zero and the formula cannot divide by it. In that case, inspect the distribution and choose another method.

## Step 6 Review before treatment

```
flags = df[df[["iqr_outlier", "z_outlier",
               "modified_z_outlier"]].any(axis=1)]

print(flags[["value", "iqr_outlier",
             "z_outlier", "modified_z_outlier"]])
```

Review the flagged rows against the source file, units, and business or scientific context. If a value is correct, keeping it may be the right choice. If it is a confirmed error, correct it from the source or mark it as missing when the true value is unknown.

## Step 7 Keep any treatment reversible

```
df["value_raw"] = df["value"]
df["value_capped"] = df["value"].clip(lower, upper)

comparison = df[["value_raw", "value_capped",
                 "iqr_outlier", "modified_z_outlier"]]
```

Capping is only an example. It should not be the automatic answer. The important design choice is to keep `value_raw` unchanged and compare results before and after any treatment.

# 5 Challenges and Gotchas

## Skewed data

A long right or left tail can be valid. A standard IQR rule may flag some of that tail. Check a histogram or boxplot and consider a transformation or a skewness-adjusted boxplot before calling those values errors.

## Masking

Several extreme values can pull the mean and standard deviation toward themselves. Their Z scores then look less extreme. This is called masking. A modified Z score or another resistant method can reveal the issue.

## Small samples

With very little data, quartiles and standard deviations can change sharply when one value changes. Use plots, source checks, and domain limits. Do not rely on one threshold alone.

## Time order and groups

A value can look unusual only because two groups or time periods were mixed. Check categories, dates, and operating conditions. For sensor or financial data, compare a value with nearby observations rather than treating every row as independent.

## Data leakage

In machine learning, calculate quartiles, means, caps, and transformation settings from the training data only. Reusing information from the validation or test set makes model performance look better than it really is.

## Many variables

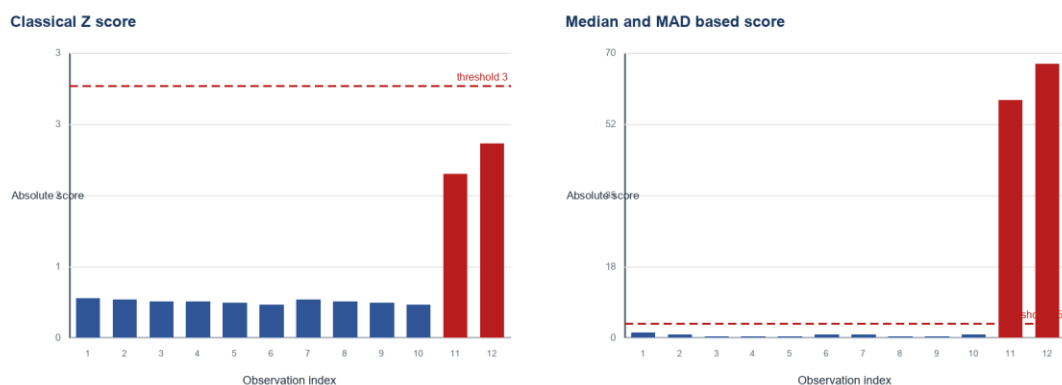
A row may look normal in each column but unusual when the columns are considered together. For multivariable data, move beyond separate column rules and consider robust distance methods or a model such as Isolation Forest.

# 6 Results and Next Steps

The example shows why method choice matters. The IQR and modified Z methods flag 180 and 200. The classical Z rule flags neither value because the two extremes increase its mean and standard deviation.

| Method            | Number flagged | Flagged values |
|-------------------|----------------|----------------|
| IQR               | 2              | 180 and 200    |
| Classical Z score | 0              | None           |
| Modified Z score  | 2              | 180 and 200    |

### How classical estimates can mask multiple extremes



Observations 11 and 12 are 180 and 200; the other ten values lie between 48 and 52.

*Two extreme values can hide each other from a classical Z score*

## What to do next

1. Run the checks on one real numeric column and inspect every flagged row.
2. Keep the raw value and record the reason for every correction or treatment.
3. Compare the analysis with and without the treatment before deciding which result to report.
4. For several variables or time series data, use a method that understands those relationships.

The main lesson is simple. Use statistics to find values that need attention, then use evidence to decide what those values mean. A clean-looking dataset is not useful if valid information was removed.

**Your turn** Share your feedback in the comments or connect with the authors on LinkedIn.

## 7 References

1. What are outliers in the data. [NIST Engineering Statistics Handbook](#)
2. Detection of outliers. [NIST Engineering Statistics Handbook](#)
3. An adjusted boxplot for skewed distributions. [Computational Statistics and Data Analysis](#)
4. Anomaly detection by robust statistics. [Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery](#)
5. Pandas Series quantile documentation. [pandas documentation](#)
6. Outlier and novelty detection. [scikit-learn documentation](#)